Assignment 3

Que 1 Write LaTeX code to produce the following expression:

$$\overbrace{a+b+c}^{\text{First Group}} + \underbrace{x+y+z}_{\text{Second Group}} = \overbrace{\underbrace{a+b}_{\text{Sum 1}}+c}^{\text{Total}}$$

Que 2 Create the following array using the `array` environment:

$$A = \left[\begin{array}{c|ccc} & x_1 & x_2 & x_3 \\ \hline y_1 & a_{11} & a_{12} & a_{13} \\ y_2 & a_{21} & a_{22} & a_{23} \\ y_3 & a_{31} & a_{32} & a_{33} \end{array}\right]$$

Que 3 Write the following piecewise function using the `cases` environment:

$$f(x) = \begin{cases} x^2 + 2x + 1, & x < 0, \\ \sqrt{x} + 1, & 0 \le x < 4, \\ \dfrac{x^2 - 1}{x - 1}, & x \ge 4. \end{cases}$$

Que 4 Typeset the following matrix and label its different sections using `\overbrace`:

$$\begin{bmatrix} \overbrace{a_{11} & a_{12} & a_{13}}^{\text{Column Group 1}} & \overbrace{a_{14} & a_{15}}^{\text{Column Group 2}} \\ a_{21} & a_{22} & a_{23} & a_{24} & a_{25} \\ a_{31} & a_{32} & a_{33} & a_{34} & a_{35} \end{bmatrix}$$

Que 5 Create the following expression

$$F(x) = \underbrace{\overbrace{\sum_{i=1}^{n} \left(a_i x_i^2 + b_i x_i + c_i\right)}^{\text{Quadratic Contribution}} + \overbrace{\begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix} A \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}}^{\text{Matrix Contribution}}}_{\text{Total Energy}}$$

where

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}.$$

Que 6 Create the following table with proper alignment, borders, and headings.

Roll No.	**Name**	**Department**	**Semester**	**CGPA**
101	Amit	CSE	2	8.45
102	Neha	CSE	2	9.12
103	Rahul	ECE	4	8.76
104	Priya	ECE	4	9.01

Requirements:

- Center-align numerical data.
- Use horizontal and vertical lines.
- Make the heading bold.

Que 7 Create the following table.

Academic Performance		
Student	**Midterm**	**Final**
A	78	89
B	85	92
C	91	94

The heading "Academic Performance" must span all three columns using `\multicolumn`.

Que 8 Create the following table.

Department	**Course**	**Credits**	**Grade**
CSE	Data Structures	4	A
	Machine Learning	4	A+
	Operating Systems	3	B+
ECE	Digital Electronics	4	A
	Signals	3	B+

Use `\multirow` so that CSE and ECE each occupy multiple rows.

Que 9 Create the following scientific table.

Model	**Parameters**	**FLOPs**	**Accuracy**	**Energy (kWh)**
ResNet18	11.7M	1.8G	94.2%	0.00093
MobileNetV2	3.5M	0.3G	92.8%	0.00061
EfficientNet-B0	5.3M	0.39G	93.5%	0.00072

Use `xcolor` and `colortbl`.

Que 10 Create a student-performance table.

Student	**Mathematics**	**Physics**	**Computer Science**
Amit	92	81	95
Neha	72	88	91
Rahul	68	74	79
Priya	96	94	98

Color Rules:

- Marks ≥ 90: Green
- Marks 75–89: Yellow
- Marks < 75: Red

Que 11 Create a table containing paragraphs inside cells.

Technique	**Description**	**Advantages**	**Limitations**
Pruning	Removes unnecessary parameters from a neural network.	Reduces model size and computation.	May reduce accuracy.
Quantization	Represents weights using lower precision.	Reduces memory and energy consumption.	Can introduce numerical error.
Knowledge Distillation	Transfers knowledge from a large teacher model to a smaller student model.	Produces compact models.	Requires training a teacher model.

Requirements:

- Use fixed-width columns.
- Allow automatic text wrapping.
- Properly align text.
- Make the table fit within the page.

Que 12 Create a professional table using the `booktabs` package.

Model	**Parameters**	**FLOPs**	**Accuracy**	**Energy**
ResNet18	11.7M	1.82G	94.21	0.00093
MobileNetV2	3.5M	0.30G	92.84	0.00061
EfficientNet-B0	5.3M	0.39G	93.52	0.00072

Requirements:

- Use `\toprule`.
- Use `\midrule`.
- Use `\bottomrule`.
- Do not use vertical lines.

- Add a caption and label.

Que 13 Create a table where one cell contains another table.

Model	**Architecture**	**Performance**	
CNN	Conv → ReLU → Pool	Metric	Value
		Accuracy	94%
		F1 Score	93%
ResNet	Conv → BN → ReLU	Metric	Value
		Accuracy	96%
		F1 Score	95%
ViT	Attention → MLP	Metric	Value
		Accuracy	97%
		F1 Score	96%

Que 14 Create the following table containing mathematical expressions.

Model	**Loss Function**	**Parameters**	**Complexity**
Linear	$L = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$	10K	$O(n)$
CNN	$L = -\sum y \log(\hat{y})$	1.2M	$O(n^2)$
Transformer	$L = -\sum_{i} y_i \log p_i$	110M	$O(n^2 d)$

— End of Assignment —